Linear Algebra — Lecture 2

Field, Vector Space, Subspace, Linear Independence, Basis and Dimension

Friedberg–Insel–Spence, *Linear Algebra* 4e, §1.2, §1.3, §1.5, §1.6
D. C. Lay, *Linear Algebra and Its Applications* 4e, §1.7, §4.1–§4.5

Topics. Field, vector space, subspace, linear independence, basis and dimension.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; Lay supplies concrete examples in $\mathbb{R}^n$ and the coordinate intuition. Mapping for this lecture:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Field, vector space	§4.1 (p. 190)	§1.2 (p. 6)
Subspace	§4.1 (p. 193), §4.2 (p. 198)	§1.3 (p. 16)
Linear independence	§1.7 (p. 55), §4.3 (p. 208)	§1.5 (p. 35)
Basis	§4.3 (p. 209), §4.4 (p. 216)	§1.6 (p. 42)
Dimension	§4.5 (p. 225)	§1.6 (p. 42–57)

Two questions that drive the whole lecture: (1) *Independence* — is there a redundant vector in the set? (2) *Spanning* — does the set have enough vectors to reach every vector of the space?

How to read: use Lay's concrete examples in $\mathbb{R}^n$ to build the picture, then Friedberg's axiomatic proofs to see why each test is legitimate.

All page numbers here are printed pages. For Lay, PDF page = printed page +17; for Friedberg, PDF page = printed page +15.

1 Fields and Vector Spaces

Definition 1.1 (Field)**.** A **field** is a set $\mathbb{F}$ with two operations, addition and multiplication, such that addition is commutative and associative with an additive identity and additive inverses, multiplication is commutative and associative with a multiplicative identity and inverses for nonzero elements, and multiplication distributes over addition.

$\mathbb{R}$ and $\mathbb{C}$ are the fields used most often. The axiomatic treatment of fields is in Friedberg, Appendix C.

Definition 1.2 (Vector space; Friedberg, §1.2, p. 6–7)**.** Let $\mathbb{F}$ be a field. A **vector space** (or linear space) over $\mathbb{F}$ is a nonempty set V with two operations, **addition** and **scalar multiplication**, satisfying the following axioms for all $x, y, z \in V$ and all $a, b \in \mathbb{F}$:

(VS 1) $x + y = y + x$ (commutativity of addition);
(VS 2) $(x + y) + z = x + (y + z)$ (associativity of addition);
(VS 3) there exists 0 with $x + 0 = x$ (additive identity);
(VS 4) for each x there exists y with $x + y = 0$ (additive inverse);
(VS 5) $1x = x$ (identity $1 \in \mathbb{F}$);
(VS 6) $(ab)x = a(bx)$;
(VS 7) $a(x + y) = ax + ay$;
(VS 8) $(a + b)x = ax + bx$.

Remark 1.1 (The two books list the axioms differently, but equivalently)**.** Lay (§4.1, p. 190) defines a vector space as a nonempty set with two operations satisfying **ten** axioms. His ten

axioms are the eight above **plus two closure requirements** ($x + y \in V$ and $ax \in V$, which he numbers Axiom 1 and Axiom 6).

The two formulations are equivalent: saying that addition is *an operation on* V already presupposes closure. Either formulation is fine to use, but cite it precisely – Lay's Axiom 3 or Friedberg's (VS 3).

Proposition 1.1 (Three facts that follow from the axioms; Lay, §4.1, p. 191, equations (1)–(3); Friedberg leaves these as exercises)**.** *For every $x \in V$ and $a \in \mathbb{F}$,*

$$0x = \mathbf{0}, \qquad a\mathbf{0} = \mathbf{0}, \qquad (-1)x = -x.$$

Proof. All three are proved the same way: split 0 as $0+0$ (or read the distributive law backwards), then cancel what is left using the zero vector and additive inverses.

(i) $0x = \mathbf{0}$**.** By (VS 8) and (VS 3),

$$0x + 0x = (0 + 0)x = 0x = 0x + \mathbf{0}.$$

Adding $-(0x)$ to both sides gives $0x = \mathbf{0}$.

(ii) $a\mathbf{0} = \mathbf{0}$**.** Likewise $a\mathbf{0} + a\mathbf{0} = a(\mathbf{0} + \mathbf{0}) = a\mathbf{0} = a\mathbf{0} + \mathbf{0}$, so $a\mathbf{0} = \mathbf{0}$.

(iii) $(-1)x = -x$**.** By (VS 8) and (i),

$$x + (-1)x = (1 + (-1))x = 0x = \mathbf{0}.$$

The vector y with $x + y = \mathbf{0}$ is unique (uniqueness of the zero vector and of inverses, Friedberg, Theorem 1.1, p. 11), so $(-1)x = -x$.

The point: none of the three is obvious. Each has to come out of the axioms, and only the zero vector, inverses and distributivity are used. □

Remark 1.2 (Examples: vector spaces are far more general than $\mathbb{R}^n$)**.**

- $\mathbb{R}^n$ for every $n \geq 1$; more generally $\mathbb{F}^n$. Lay Example 1, p. 191.
- All directed line segments in space (equal when same length and direction), with the parallelogram rule. Lay Example 2, p. 191.
- All doubly infinite sequences $\{y_k\} = (\dots, y_{-1}, y_0, y_1, \dots)$. Lay Example 3, p. 191.
- $\mathcal{F}(S, \mathbb{F})$: for a nonempty set S and a field $\mathbb{F}$, **all functions** from S to $\mathbb{F}$, with addition and scalar multiplication defined **pointwise**: $(f + g)(s) = f(s) + g(s)$ and $(cf)(s) = c\,f(s)$. Friedberg Example 3, §1.2, p. 9.
- $P(\mathbb{F})$: **all** polynomials with coefficients in $\mathbb{F}$, of any degree. Lay §4.1, p. 193; Friedberg Example 4, §1.2, p. 10.
- $P_n(\mathbb{F})$: polynomials of degree at most n with coefficients in $\mathbb{F}$. Lay §4.1, p. 192; Friedberg Example 5, §1.2, p. 11.
- $M_{m\times n}(\mathbb{F})$: $m \times n$ matrices with the usual operations. Friedberg Example 2, §1.2, p. 9.

The point: a "vector" need not be an arrow or a tuple; only the axioms matter.

2 Subspaces

Definition 2.1 (Subspace; Lay, §4.1, p. 193, parts a–c; Friedberg, Theorem 1.3, p. 17)**.** Let W be a nonempty subset of a vector space V. Then W is a **subspace** of V if W is itself a vector space under the operations of V. Equivalently (this is Theorem 2.1 below), only three conditions need checking:

(a) $\mathbf{0} \in W$;
(b) $x + y \in W$ whenever $x, y \in W$ (closed under addition);

(c) $ax \in W$ whenever $a \in \mathbb{F}$ and $x \in W$ (closed under scalar multiplication).

Theorem 2.1 (Criterion for a subspace; Friedberg, Theorem 1.3, §1.3, p. 17)**.** *Let V be a vector space and $W \subseteq V$. Then W is a subspace of V if and only if conditions (a), (b), (c) above hold for the operations defined on V.*

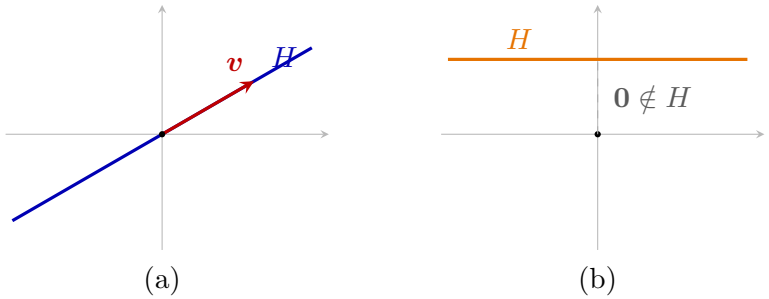
Proof. **($\Rightarrow$)** If W is a subspace then W is a vector space under the operations of V, so (b) and (c) hold. Also W has its own zero $0'$, that is, $x + 0' = x$ for every $x \in W$; but $x + 0 = x$ holds as well, so $0' = 0$ by uniqueness of the zero vector (Friedberg, Theorem 1.1, p. 11) and (a) holds.

($\Leftarrow$) Assume (a), (b), (c). By (b) and (c) W is closed under both operations, so they restrict to W. Of the eight axioms, commutativity, associativity, the unit law $1x = x$, $a(bx) = (ab)x$ and the two distributive laws hold for every vector of V and hence for every vector of W; (a) supplies the zero axiom. Only the additive inverse is left: if $x \in W$ then $(-1)x \in W$ by (c), and $(-1)x = -x$ (Proposition 1.1), so $-x \in W$. Hence W satisfies all eight axioms, is a vector space, and therefore a subspace of V. □

Theorem 2.2 (Intersections of subspaces; Friedberg, Theorem 1.4, §1.3, p. 19)**.** *Any intersection of subspaces is again a subspace: if W_1 and W_2 are subspaces of V, then so is $W_1 \cap W_2$.*

*Caution: the **union** of subspaces is usually **not** a subspace. Two distinct lines through the origin in $\mathbb{R}^2$, for instance, fail to be closed under addition.*

Remark 2.1 (Geometric picture)**.** In $\mathbb{R}^3$, every line through the origin and every plane through the origin is a subspace. Failing to pass through the origin is enough to destroy the property, because then the zero vector is missing and condition (a) fails immediately.



(a) a line through the origin is a subspace of $\mathbb{R}^3$; (b) a line not through the origin is not a subspace — it does not contain the zero vector.

Theorem 2.3 (A span is a subspace; Lay, Theorem 1, §4.1, p. 194; this is the same statement as Friedberg's Theorem 1.5 from Lecture 1)**.** *If $v_1, \dots, v_p$ lie in a vector space V, then* $\operatorname{Span}\{v_1, \dots, v_p\}$ *is a **subspace** of V.*

Definition 2.2 (Null space; Lay, §4.2, p. 199)**.** Let A be an $m \times n$ matrix. The **null space** of A, written $\operatorname{Nul} A$, is the set of all solutions of the homogeneous equation $A\boldsymbol{x} = \mathbf{0}$:

$$\operatorname{Nul} A = \{\boldsymbol{x} \in \mathbb{R}^n : A\boldsymbol{x} = \mathbf{0}\}.$$

Theorem 2.4 (The null space is a subspace; Lay, Theorem 2, §4.2, p. 199)**.** *The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.*

Definition 2.3 (Column space; Lay, §4.2, p. 201)**.** The **column space** of an $m \times n$ matrix A, written $\operatorname{Col} A$, is the set of all linear combinations of the columns of A. If $A = [\,\boldsymbol{a}_1 \ \cdots \ \boldsymbol{a}_n\,]$, then $\operatorname{Col} A = \operatorname{Span}\{\boldsymbol{a}_1, \dots, \boldsymbol{a}_n\}$.

Theorem 2.5 (The column space is a subspace; Lay, Theorem 3, §4.2, p. 201)**.** *The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$, and a typical vector in it can be written as $A\boldsymbol{x}$ for some $\boldsymbol{x}$.*

3 Linear Dependence and Independence

Definition 3.1 (Linearly independent and linearly dependent; Lay, §1.7, p. 56)**.** An indexed set $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation

$$x_1\boldsymbol{v}_1 + x_2\boldsymbol{v}_2 + \cdots + x_p\boldsymbol{v}_p = \boldsymbol{0}$$

has only the trivial solution ($x_1 = \cdots = x_p = 0$), and **linearly dependent** otherwise.

In a general vector space the phrase "only the trivial solution" must be replaced by "$c_1v_1 + \cdots + c_pv_p = 0$ forces all $c_i = 0$", because there is no system of equations to write down. Friedberg stresses this point in §1.5.

Theorem 3.1 (Characterisation of linearly dependent sets; Lay, Theorem 7, §1.7, p. 58; the general-vector-space form is Lay, Theorem 4, §4.3, p. 208)**.** *An indexed set $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ with $p \geq 2$ and $\boldsymbol{v}_1 \neq \boldsymbol{0}$ is linearly dependent if and only if some $\boldsymbol{v}_j$ ($j > 1$) is a linear combination of the preceding vectors $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{j-1}$.*

Proof. **(⇐)** Suppose $\boldsymbol{v}_j = c_1\boldsymbol{v}_1 + \cdots + c_{j-1}\boldsymbol{v}_{j-1}$ with $j > 1$. Then

$$c_1\boldsymbol{v}_1 + \cdots + c_{j-1}\boldsymbol{v}_{j-1} + (-1)\boldsymbol{v}_j = \boldsymbol{0}$$

is a relation with a *nonzero* coefficient (the one on $\boldsymbol{v}_j$), so S is dependent.

(⇒) Suppose S is dependent, so $c_1\boldsymbol{v}_1 + \cdots + c_p\boldsymbol{v}_p = \boldsymbol{0}$ with the c_i not all zero. Take the *largest* index j with $c_j \neq 0$.

If $j = 1$ the relation collapses to $c_1\boldsymbol{v}_1 = \boldsymbol{0}$, and $\boldsymbol{v}_1 \neq \boldsymbol{0}$ forces $c_1 = 0$, a contradiction; hence $j > 1$. Every coefficient beyond j is zero, so the relation only involves $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_j$:

$$c_1\boldsymbol{v}_1 + \cdots + c_j\boldsymbol{v}_j = \boldsymbol{0} \quad \Longrightarrow \quad \boldsymbol{v}_j = -\frac{c_1}{c_j}\boldsymbol{v}_1 - \cdots - \frac{c_{j-1}}{c_j}\boldsymbol{v}_{j-1},$$

that is, $\boldsymbol{v}_j$ is a combination of the vectors before it.

The one idea is taking the *largest* such j: it kills every coefficient after j, which is what leaves a $\boldsymbol{v}_j$ that can be solved for. Note also that the theorem only promises *some* such j – not every vector of a dependent set is a combination of the preceding ones. □

Theorem 3.2 (Two quick tests; Lay, Theorem 8, §1.7, p. 59; Lay, Theorem 9, §1.7, p. 59)**.**

*(a) If a set contains **more vectors than entries in each vector**, it is linearly dependent: any set of $p > n$ vectors in $\mathbb{R}^n$ is linearly dependent.*

*(b) If a set contains the **zero vector**, it is linearly dependent.*

Note that (a) says nothing about the case $p \leq n$; there both outcomes are possible and one has to solve.

Proof. **(a)** Let $A = [\,\boldsymbol{v}_1 \ \cdots \ \boldsymbol{v}_p\,]$. Then A is $n \times p$, and $A\boldsymbol{x} = \boldsymbol{0}$ is a homogeneous system of n equations in p unknowns. If $p > n$ there are more unknowns than equations, so there are at most n pivot columns and hence a free variable; therefore $A\boldsymbol{x} = \boldsymbol{0}$ has a nontrivial solution: $c_1\boldsymbol{v}_1 + \cdots + c_p\boldsymbol{v}_p = \boldsymbol{0}$ with the c_i not all zero, so S is dependent.

(b) Suppose $\boldsymbol{v}_k = \boldsymbol{0}$. Take $c_k = 1$ and every other $c_i = 0$; then $c_1\boldsymbol{v}_1 + \cdots + c_p\boldsymbol{v}_p = \boldsymbol{0}$ with coefficients not all zero, so S is dependent.

Both are the same move – assume independence, then make it contradict itself: with $\mathbf{0}$ in the set, the "trivial solution only" that independence demands is broken by $1 \cdot \mathbf{0} = \mathbf{0}$. □

Theorem 3.3 (Monotonicity; Friedberg, Theorem 1.6 and its corollary, §1.5, p. 39)**.** *Let V be a vector space and $S_1 \subseteq S_2 \subseteq V$.*

(a) If S_1 is linearly dependent, then S_2 is linearly dependent.
(b) (Corollary) If S_2 is linearly independent, then S_1 is linearly independent.

In words: growing a set cannot destroy dependence, shrinking it cannot destroy independence.

Theorem 3.4 (When can one vector be added?; Friedberg, Theorem 1.7, §1.5, p. 39)**.** *Let S be a linearly independent subset of a vector space V and let $v \in V$ with $v \notin S$. Then*

$$S \cup \{v\} \text{ is linearly dependent} \iff v \in \operatorname{Span}(S).$$

Proof. **(⇒)** Suppose $S \cup \{\boldsymbol{v}\}$ is dependent, so there are coefficients, not all zero, with

$$a\boldsymbol{v} + a_1\boldsymbol{u}_1 + \cdots + a_k\boldsymbol{u}_k = \mathbf{0}, \qquad \boldsymbol{u}_i \in S.$$

Here $a \neq 0$: if a were zero we would have a nontrivial relation inside S, contradicting its independence. Hence

$$\boldsymbol{v} = -\frac{a_1}{a}\boldsymbol{u}_1 - \cdots - \frac{a_k}{a}\boldsymbol{u}_k \in \operatorname{Span}(S).$$

(⇐) Suppose $\boldsymbol{v} \in \operatorname{Span}(S)$, say $\boldsymbol{v} = b_1\boldsymbol{v}_1 + \cdots + b_m\boldsymbol{v}_m$ with $\boldsymbol{v}_i \in S$. Then

$$b_1\boldsymbol{v}_1 + \cdots + b_m\boldsymbol{v}_m + (-1)\boldsymbol{v} = \mathbf{0}$$

has coefficients not all zero (the one on $\boldsymbol{v}$ is -1, and $\boldsymbol{v} \notin S$ makes it distinct from the $\boldsymbol{v}_i$), so $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{v}\}$ is dependent, and so is the larger set $S \cup \{\boldsymbol{v}\}$ (Theorem 3.3(a)).

This criterion is the tool the Replacement Theorem leans on: it turns "$\boldsymbol{v}$ lies in the span" into "adjoining $\boldsymbol{v}$ to an independent set destroys independence". □

4 Bases

Definition 4.1 (Basis; Lay, §4.3, p. 209)**.** Let H be a subspace of a vector space V. An indexed set $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_p\}$ in V is a **basis** for H if (i) B is linearly independent, and (ii) $H = \operatorname{Span}\{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_p\}$.

Taking $H = V$ gives a basis of V: a linearly independent set that spans V. Informally, a basis is a set that is *neither too small nor too large* – exactly enough.

Theorem 4.1 (Unique representation; Lay, Theorem 7, the Unique Representation Theorem, §4.4, p. 216; Friedberg, Theorem 1.8, §1.6, p. 43)**.** *Let $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\}$ be a basis for a vector space V. Then for each $\boldsymbol{x} \in V$ there is a **unique** choice of scalars $c_1, \ldots, c_n$ with*

$$\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n.$$

Existence comes from "B spans V"; uniqueness from "B is linearly independent" – subtracting two representations yields a linear relation on B, whose coefficients must all vanish.

Theorem 4.2 (Spanning Set Theorem; Lay, Theorem 5, §4.3, p. 210)**.** *Let $S = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p\}$ be a set in V and let $H = \operatorname{Span} S$.*

(a) If some vector in S is a linear combination of the remaining vectors, then removing it still leaves a set that spans H.

(b) If $H \neq \{\mathbf{0}\}$, then some subset of S is a basis for H.

Theorem 4.3 (Finitely generated implies a finite basis; Friedberg, Theorem 1.9, §1.6, p. 44)**.** *If a vector space V is generated by a finite set S, then some subset of S is a basis for V; in particular V has a finite basis.*

Definition 4.2 (Coordinate vector; Lay, §4.4, p. 216)**.** Let $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\}$ be a basis for V and let $\boldsymbol{x} \in V$. The weights $c_1, \ldots, c_n$ with $\boldsymbol{x} = c_1\boldsymbol{b}_1 + \cdots + c_n\boldsymbol{b}_n$ are the **coordinates of $\boldsymbol{x}$ relative to** B, and

$$[\boldsymbol{x}]_B = \begin{bmatrix} c_1 \\ \vdots \\ c_n \end{bmatrix} \in \mathbb{R}^n$$

is the **coordinate vector** of $\boldsymbol{x}$ relative to B.

Theorem 4.4 (The coordinate mapping is an isomorphism; Lay, Theorem 8, §4.4, p. 219)**.** *If $B = \{\boldsymbol{b}_1, \ldots, \boldsymbol{b}_n\}$ is a basis for V, then the coordinate mapping $\boldsymbol{x} \mapsto [\boldsymbol{x}]_B$ is a **one-to-one** linear transformation from V onto $\mathbb{R}^n$.*

Theorem 4.5 (Replacement Theorem; Friedberg, Theorem 1.10, §1.6, p. 45)**.** *Let V be a vector space generated by a set G containing exactly n vectors, and let L be a linearly independent subset of V containing exactly m vectors. Then $m \leq n$ and there is a subset H of G containing exactly $n - m$ vectors such that $L \cup H$ generates V.*

Proof. By induction on m.

$m = 0$**:** here $L = \varnothing$, and $H = G$ does the job ($|H| = n = n - 0$).

Induction step: assume the theorem for some $m \geq 0$ and prove it for $m + 1$. Let $L = \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_{m+1}\}$ be independent. By Theorem 3.3(b) the subset $\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$ is independent, so the induction hypothesis gives $m \leq n$ and a subset $\{\boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-m}\}$ of G such that

$$B := \{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{u}_1, \ldots, \boldsymbol{u}_{n-m}\}$$

generates V. Hence $\boldsymbol{v}_{m+1}$ is a combination of the elements of B:

$$a_1\boldsymbol{v}_1 + \cdots + a_m\boldsymbol{v}_m + b_1\boldsymbol{u}_1 + \cdots + b_{n-m}\boldsymbol{u}_{n-m} = \boldsymbol{v}_{m+1}. \qquad (*)$$

If $n - m = 0$, the right-hand side of $(*)$ is a combination of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$, i.e. $\boldsymbol{v}_{m+1} \in \mathrm{Span}\{\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m\}$, which contradicts the independence of L by Theorem 3.4. So $n - m > 0$. By the same argument at least one $b_i \neq 0$ – otherwise $\boldsymbol{v}_{m+1}$ would again be a combination of $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m$ – so assume $b_1 \neq 0$ and solve $(*)$ for $\boldsymbol{u}_1$:

$$\boldsymbol{u}_1 = -\frac{a_1}{b_1}\boldsymbol{v}_1 - \cdots - \frac{a_m}{b_1}\boldsymbol{v}_m + \frac{1}{b_1}\boldsymbol{v}_{m+1} - \frac{b_2}{b_1}\boldsymbol{u}_2 - \cdots - \frac{b_{n-m}}{b_1}\boldsymbol{u}_{n-m}.$$

Take $H = \{\boldsymbol{u}_2, \ldots, \boldsymbol{u}_{n-m}\}$. The display above says $\boldsymbol{u}_1 \in \mathrm{Span}(L \cup H)$; and

$$\boldsymbol{v}_1, \ldots, \boldsymbol{v}_m, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{n-m} \in \mathrm{Span}(L \cup H),$$

so $B \subseteq \mathrm{Span}(L \cup H)$. Since B generates V, $V = \mathrm{Span}(B) \subseteq \mathrm{Span}(L \cup H) \subseteq V$ (Theorem 2.3, which is Friedberg's Theorem 1.5: Span is the smallest subspace containing $L \cup H$), so $L \cup H$ generates V. Finally $H \subseteq G$ and $|H| = (n - m) - 1 = n - (m + 1)$, which completes the induction. □

Theorem 4.6 (Two corollaries of the Replacement Theorem; Friedberg, §1.6, Corollary 1 (p. 46) and Corollary 2 (p. 47))**.**

*(a) If V has a finite basis, then **every basis of V contains the same number of vectors**.*

(b) Let $\dim V = n$*. Any finite generating set for* V *contains at least* n *vectors, and a generating set containing exactly* n *vectors is a basis.*

Proof. **(a)** Let β be a finite basis with n vectors and let γ be any basis. If $|\gamma| > n$, apply the Replacement Theorem with $G = \beta$ (n vectors) and L equal to any $n+1$ vectors of γ: it gives $n+1 \le n$, a contradiction, so $|\gamma| \le n$. Swapping the roles ($G = \gamma$, $L = \beta$) gives $n \le |\gamma|$. Hence $|\gamma| = n$: the size of a basis does not depend on which basis you pick, and that invariant is the **dimension**.

(b) First, "at least". Let G be a finite generating set with m vectors and let β be a basis with n vectors. Apply the Replacement Theorem with G (m vectors, generating) and $L = \beta$ (n vectors, independent): $n \le m$.

Now "exactly". Let $m = n$. By Theorem 4.3 some subset β' of G is a basis; every basis has n vectors by (a), so $|\beta'| = n$; and the only n-element subset of an n-element set is the set itself, so $\beta' = G$. Thus G is a basis. □

5 Dimension

Definition 5.1 (Dimension; Lay, §4.5, p. 226)**.** If V is spanned by a finite set, then V is **finite-dimensional** and its **dimension** $\dim V$ is the number of vectors in a basis for V; the dimension of the zero space $\{\mathbf{0}\}$ is defined to be 0. If V cannot be spanned by a finite set, it is **infinite-dimensional**.

The definition is legitimate precisely because of Theorem 4.6(a): the size of a basis does not depend on which basis is chosen.

Theorem 5.1 (Two basic facts about dimension; Lay, Theorem 9, §4.5, p. 225; Lay, Theorem 10, §4.5, p. 226)**.**

(a) If a basis of V *contains* n *vectors, then any set in* V *containing more than* n *vectors must be linearly dependent.*

(b) If V *has a basis of* n *vectors, then every basis of* V *consists of exactly* n *vectors.*

Proof. These are the same facts as Theorem 4.6, phrased the way Lay states them; one application of the Replacement Theorem does it.

(a) Let β be a basis with n vectors and let S have $p > n$ vectors. If S were independent, the Replacement Theorem applied to the generating set β (n vectors) and $L = S$ (p vectors) would give $p \le n$, contradicting $p > n$. So S is dependent.

(b) This is Theorem 4.6(a): every basis of V has exactly n vectors.

(Read (a) backwards for the everyday use: to show a set is dependent, it suffices to count – if it has more vectors than the dimension, it must be dependent.) □

Theorem 5.2 (A subspace is no larger than the whole space; Friedberg, §1.6, corollary on p. 51; Lay, Theorem 11, §4.5, p. 227; Friedberg, Theorem 1.11, §1.6, p. 50)**.** *Let* W *be a subspace of a finite-dimensional vector space* V*. Then* W *is also finite-dimensional and* $\dim W \le \dim V$*; moreover, if* $\dim W = \dim V$ *then* $W = V$*.*

The corresponding extension statement: any basis for W *can be extended to a basis for* V*.*

Proof. Let $\dim V = n$.

If $W = \{\mathbf{0}\}$ then W is finite-dimensional and $\dim W = 0 \le n$. Otherwise choose vectors $\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots$ in W one at a time, keeping the set independent at every step. No independent subset of V has more than n vectors (Theorem 5.1(a)), so the process stops at some $k \le n$: $\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k\}$ is

independent, while adjoining any further vector of W would make it dependent. By Theorem 3.4, every vector of W lies in $\mathrm{Span}\{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_k\}$, so that set generates W and is a basis of W; hence $\dim W = k \leq n$.

If $\dim W = n$: a basis of W is an independent subset of V with n vectors. If it did not generate V there would be $\boldsymbol{v} \in V \setminus \mathrm{Span}(W)$, and by Theorem 3.4 adjoining $\boldsymbol{v}$ would give an independent set with $n+1$ vectors, contradicting Theorem 5.1(a). So it generates V, i.e. $W = \mathrm{Span}(W) = V$. □

Theorem 5.3 (Basis Theorem; Lay, Theorem 12, the Basis Theorem, §4.5, p. 227)**.** *Let V be a p-dimensional vector space with $p \geq 1$. Then*

(a) any linearly independent set of exactly p vectors in V is automatically a basis for V;
(b) any set of exactly p vectors that spans V is automatically a basis for V.

This is what makes verification cheap: once the dimension is known, only one of the two defining conditions needs checking.

Proof. **(a)** Let S have p vectors and be independent. If S did not generate V there would be $\boldsymbol{v} \in V \setminus \mathrm{Span}(S)$; by Theorem 3.4 the set $S \cup \{\boldsymbol{v}\}$ would be independent, giving an independent set of $p+1$ vectors and contradicting Theorem 5.1(a). So S generates V and is a basis.

(b) Let S have p vectors and generate V. This is Theorem 4.6(b) with $n = p$: a generating set with exactly n vectors is a basis. So S is a basis.

The theorem halves the work of verifying a basis: once the dimension is known, only *one* of independence and spanning has to be checked. □

6 A Worked Example

Example 6.1 (A subspace of $\mathbb{R}^3$, its basis and its dimension; This example follows the discussion in Lay, §4.3, p. 208, and Lay, §4.5, Examples 1–2)**.** Let

$$H = \left\{ \begin{bmatrix} a \\ b \\ c \end{bmatrix} \in \mathbb{R}^3 : a - 3b - c = 0 \right\}.$$

Decide whether H is a subspace of $\mathbb{R}^3$, find a basis and the dimension of H, and decide whether $\boldsymbol{w} = (4, 0, 1)^\top$ belongs to H.

Solution. **(1) Is H a subspace of $\mathbb{R}^3$?** Rewrite the constraint as $a = 3b + c$. Then

$$\begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 3b + c \\ b \\ c \end{bmatrix} = b \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix},$$

so $H = \mathrm{Span}\left\{ \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$. By Theorem 2.3 a span is a subspace, hence so is H.

(2) Find a basis and the dimension. The two vectors are not proportional, hence linearly independent, so they form a basis of H and $\dim H = 2$.

(3) Is $\boldsymbol{w} = (4, 0, 1)^\top$ in H? $4 - 3 \cdot 0 - 1 = 3 \neq 0$, so $\boldsymbol{w} \notin H$. Equivalently, with $b = 0$ and $c = 1$ we get $3b + c = 1 \neq 4$: the first coordinate 4 is not obtainable from $(3, 1)$ and $(1, 0)$. □

Approach. When H is given by a single linear constraint, first solve the constraint for the free parameters: once $H = \mathrm{Span}\{\cdots\}$ is written down, the subspace conclusion follows immediately from the spanning-set theorem, the parameter vectors are a spanning set that one proportionality check promotes to a basis, and their number is the dimension.

7 Supplement: notions Homework 2 uses but the lecture did not cover

Remark 7.1 (Where these come from)**.** None of the following is part of the lecture's main thread, yet Homework 2 uses them:

- The **sum** of subsets and the **direct sum**: the **exercises** of Friedberg, §1.3 (printed p. 22). The homework sheet supplies its own definitions (Definitions 1 and 2 there).
- **Cosets** and the **quotient space**: Friedberg, §1.3 Exercise 31 (printed p. 23). The sheet gives **no** definition, so it has to be looked up in the book — this is Homework 2 Problem 11 (20 points).
- The **trace**: Friedberg, §1.3 Example 3 (printed p. 18).
- The **matrix units** E^{ij}: Friedberg, §1.5 Exercise 6 (printed p. 41).

Definition 7.1 (Sum of subsets, direct sum; Friedberg, §1.3 exercises, p. 22)**.** If S_1, S_2 are nonempty subsets of a vector space V, their **sum** is $S_1 + S_2 = \{x + y : x \in S_1,\ y \in S_2\}$. If W_1, W_2 are subspaces of V with $W_1 \cap W_2 = \{0\}$ and $W_1 + W_2 = V$, then V is the **direct sum** of W_1 and W_2, written $V = W_1 \oplus W_2$.

Proposition 7.1 (Dimension of a sum; Friedberg §1.6, Exercise 29, p. 57)**.** *Let W_1, W_2 be finite-dimensional subspaces of a vector space V. Then $W_1 + W_2$ is finite-dimensional and*

$$\dim(W_1 + W_2) = \dim W_1 + \dim W_2 - \dim(W_1 \cap W_2).$$

In particular, $\dim(W_1 + W_2) = \dim W_1 + \dim W_2$ *if and only if* $W_1 \cap W_2 = \{\mathbf{0}\}$.

Proof. Take a basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t$ of $W_1 \cap W_2$. By the Replacement Theorem and the Basis Theorem it extends to a basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t, \boldsymbol{v}_1, \dots, \boldsymbol{v}_p$ of W_1, and also to a basis $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t, \boldsymbol{w}_1, \dots, \boldsymbol{w}_q$ of W_2. Hence $\dim(W_1 \cap W_2) = t$, $\dim W_1 = t + p$ and $\dim W_2 = t + q$.

These $t+p+q$ vectors span W_1+W_2: every element of W_1 is a combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t, \boldsymbol{v}_1, \dots, \boldsymbol{v}_p$, every element of W_2 is a combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t, \boldsymbol{w}_1, \dots, \boldsymbol{w}_q$, and a sum of two such combinations is again one.

They are linearly independent. If $\sum_i a_i \boldsymbol{u}_i + \sum_j b_j \boldsymbol{v}_j + \sum_k c_k \boldsymbol{w}_k = \mathbf{0}$, then $\sum_k c_k \boldsymbol{w}_k$ lies in W_2 and equals $-\big(\sum_i a_i \boldsymbol{u}_i + \sum_j b_j \boldsymbol{v}_j\big)$, which lies in W_1; so it lies in $W_1 \cap W_2$ and is a combination of $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t$. But $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t, \boldsymbol{w}_1, \dots, \boldsymbol{w}_q$ is a basis of W_2 and a vector has unique coefficients in a basis, so every $c_k = 0$. Then $\sum_i a_i \boldsymbol{u}_i + \sum_j b_j \boldsymbol{v}_j = \mathbf{0}$, and since $\boldsymbol{u}_1, \dots, \boldsymbol{u}_t, \boldsymbol{v}_1, \dots, \boldsymbol{v}_p$ is a basis of W_1, all a_i and b_j vanish.

So these $t+p+q$ vectors form a basis of W_1+W_2 and $\dim(W_1+W_2) = t+p+q$. From $t+p = \dim W_1$ and $t + q = \dim W_2$ we get $t + p + q = \dim W_1 + \dim W_2 - t$, which is the claimed identity.

Finally, $\dim(W_1 + W_2) = \dim W_1 + \dim W_2$ holds exactly when $t = 0$, that is, when $W_1 \cap W_2 = \{\mathbf{0}\}$. □

Definition 7.2 (Coset and quotient space; Friedberg, §1.3 Exercise 31, p. 23)**.** Let W be a subspace of V and $v \in V$. The set $\{v\} + W = \{v + w : w \in W\}$ is the **coset** of W containing v, customarily written $v + W$. The set of all cosets is V/W; with

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W, \qquad a(v + W) = av + W,$$

it becomes a vector space, the **quotient space** of V modulo W.

Proposition 7.2 (Basic properties of cosets; Friedberg, §1.3 Exercise 31, p. 23)**.** *Let W be a subspace of V and $v, v_1, v_2 \in V$.*

(1) $v + W$ is a subspace of $V \iff v \in W$;

(2) $v_1 + W = v_2 + W \iff v_1 - v_2 \in W$*;*
(3) the two operations above do not depend on the representatives chosen;
(4) V/W *is a vector space under those two operations.*

Proof. (1) If $v+W$ is a subspace, then $0 \in v+W$, so $0 = v+w$ for some $w \in W$, hence $v = -w \in W$. Conversely, if $v \in W$ then $v+W \subseteq W$ (every $v+w$ lies in W) and $W \subseteq v+W$ (as $w = v+(w-v)$), so $v + W = W$ is a subspace.

(2) If $v_1+W = v_2+W$, then $v_1 \in v_1+W = v_2+W$ gives $v_1 = v_2+w$ with $w \in W$, so $v_1-v_2 = w \in W$. Conversely, if $v_1 - v_2 \in W$ then $v_1 = v_2 + (v_1 - v_2) \in v_2 + W$, so $v_1 + W \subseteq v_2 + W$; swapping the subscripts gives the reverse inclusion, and the two cosets are equal.

(3) Suppose $v_1 + W = v_1' + W$ and $v_2 + W = v_2' + W$. By (2), $v_1 - v_1'$ and $v_2 - v_2'$ lie in W, so

$$(v_1 + v_2) - (v_1' + v_2') = (v_1 - v_1') + (v_2 - v_2') \in W, \qquad av_1 - av_1' = a(v_1 - v_1') \in W,$$

and (2) again gives $(v_1 + W) + (v_2 + W) = (v_1' + W) + (v_2' + W)$ and $a(v_1 + W) = a(v_1' + W)$.

(4) Verify the eight axioms through the representatives: $0 + W$ is the zero element, $-v + W$ is the negative of $v + W$, and the remaining six follow by substituting representatives into the axioms of V, for instance

$$(v_1 + W) + (v_2 + W) = (v_1 + v_2) + W = (v_2 + v_1) + W = (v_2 + W) + (v_1 + W). \qquad \square$$

Definition 7.3 (Trace; Friedberg, §1.3 Example 3, p. 18)**.** The **trace** of an $n \times n$ matrix M is the **sum** of its diagonal entries: $\operatorname{tr}(M) = M_{11} + M_{22} + \cdots + M_{nn}$.

Definition 7.4 (Matrix units; Friedberg, §1.5 Exercise 6, p. 41)**.** In $M_{m\times n}(F)$ let E^{ij} be the matrix whose only nonzero entry is a 1 in row i and column j. The set $\{E^{ij} : 1 \le i \le m,\ 1 \le j \le n\}$ is linearly independent.

8 Checklist and Exercises

By the end of this lecture you should be able to:

1. write down the eight (or ten) axioms of a vector space and explain why the two formulations agree;
2. give an example showing that "a set with two closed operations" need not be a vector space, naming the axiom that fails;
3. decide whether a given subset of $\mathbb{R}^n$ is a subspace, and say which condition fails;
4. explain why a line not through the origin is not a subspace, using the zero vector;
5. decide whether a set of vectors is linearly dependent – both by solving and by using Theorem 3.1 and the two quick tests;
6. find a basis and the dimension of a given subspace;
7. state the Unique Representation Theorem and relate it to "basis = independent + spanning";
8. use the two corollaries of the Replacement Theorem to bound how many vectors a basis can have.

Exercises (printed pages). Lay: §1.7 p. 60, §4.1 p. 195, §4.2 p. 205, §4.3 p. 213, §4.4 p. 222, §4.5 p. 229; Friedberg: §1.2 p. 12, §1.3 p. 19, §1.5 p. 40, §1.6 p. 53. Each section also opens its exercises with Practice Problems; do those first.

How to use the two texts together:

- **Examples from Lay, theorems from Friedberg.** Lay's $\mathbb{R}^n$ examples are the most concrete; Friedberg proves everything over a general field $\mathbb{F}$ and contains the Replacement Theorem and its corollaries, which Lay does not have.

- The spine of this lecture: *axioms define the space* $\to$ *subspaces are tested by three conditions* $\to$ *a basis is "independent + spanning"* $\to$ *the dimension is the invariant size of a basis.*
- Lay's §1.7 treats linear independence in $\mathbb{R}^n$ and only §4.3 moves it to a general vector space. If time is short, read Theorem 3.1 (that is, Lay's Theorem 4 in §4.3) and the definition of a basis first, then go back to §1.7.